2.2 Probability of Receiving Tenure

We discuss the relationship between tenure and publication in journals of different quality tiers. Figure 3 plots average predicted probabilities of tenure associated with different numbers of publications in the four journal categories estimated using a logit model.[26,27] Controlling for the total number of publications in all specifications, we isolate a composition effect from a scale effect. We control for gender, number of co-authors, and the quality of the graduate school attended. Lastly, we control for the quality of authors' publication portfolios by including a vector of statistics that summarize the distribution of field-adjusted citations received by each author's portfolio of publications.[28,29]

Figure 3 shows that publishing in T5 journals is associated with the largest increases in probabilities of receiving tenure. An individual with a single T5 publication is predicted to have a 0.3 probability of receiving tenure. The predicted probability increases to 0.43 and 0.62 for individuals with two and three T5 publications respectively. Although publishing in non-T5 outlets is associated with non-zero probabilities of receiving tenure that are statistically significant at the 5% level, the predicted probabilities associated with these publications are considerably lower than those associated with T5 publications. Among the non-T5 estimates, the largest probability of receiving tenure is 0.25 and it is associated with publishing two articles in Tier A journals. This probability is lower than the probability of 0.3 associated with publishing a single T5 article. The probability of 0.62 associated with three or more

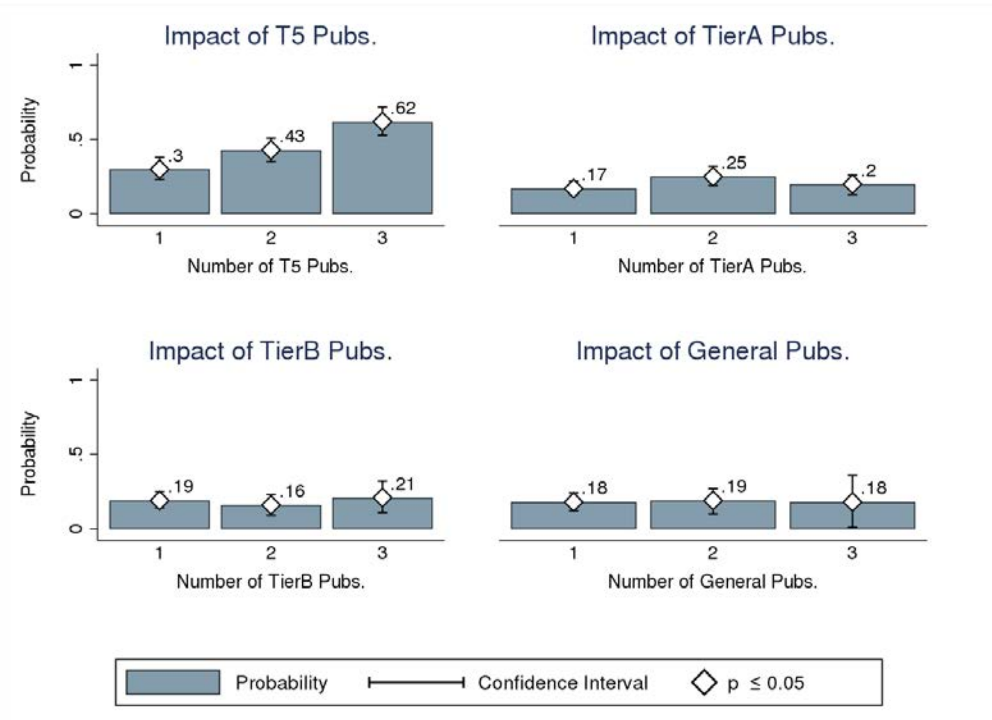
[26] See Text-Appendix Section 1.1 for the exact specification used in our Logit estimations.

[27] The corresponding marginal effects are presented under the "Pooled" columns of the Online Appendix Table O-A13. Online Appendix Table O-A10 presents comparable estimates of partial effects obtained from our Linear Probability Model (LPM) estimation. Results are qualitatively the same. The T5 remains the most influential category by far.

[28] Relevance of an article varies by analysis. The current estimates of tenure by the first spell utilize citations for all articles published during the first spell. Estimates for tenure by the 7^{th} year utilize citations to articles published by the 7^{th} year of tenure-track experience.

[29] The vector of citation controls includes the following statistics that summarize the distribution of field-adjusted citations received by each author's portfolio of publications: 25^{th} percentile, median, 75^{th} percentile, minimum, maximum, and mean field-adjusted citations. Our adjustment follows a citation-rescaling procedure similar to the one introduced by Radicchi et al. (2008) and discussed by Perry and Reny (2016). Specifically, it rescales citations received by each article i with the mean number of citations received by all articles in i's field published during i's year of publication. See Online Appendix Section 2 for detailed documentation of the procedure undertaken to adjust citations by field and year.

Figure 3: Predicted Probabilities for Tenure Receipt in the First Spell of Tenure-Track Employment (Logit)



Note: This figure plots the predicted probabilities associated with different levels of publications in different journal categories. The predicted probability is defined in Equation TA-2 (Equation TA-2 uses parameter estimates from Equation TA-1). White diamonds on the bars indicate that the prediction is significantly different than zero at the 5% level.

T5 publications is approximately 150% greater than this largest non-T5 estimate. The pattern of large differences between the probability-of-tenure associated with T5 and non-T5 publications persists when we investigate the relationship between publications and the probability of receiving tenure by the 7^{th} year of tenure-track employment.[30]

2.2.1 The Power of the T5 by Department Rank

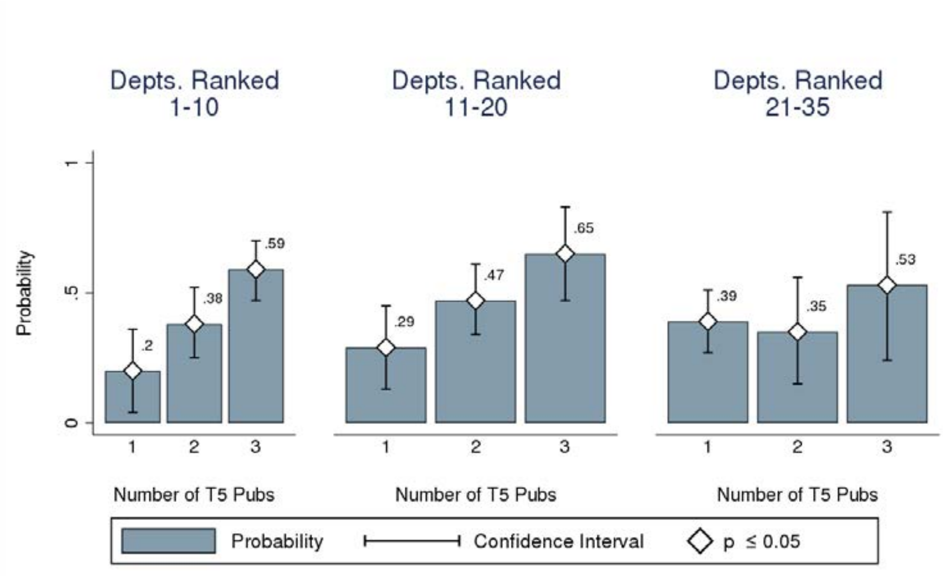
Figure 4 plots department rank-specific predicted probabilities for receipt of tenure during the first spell of tenure-track experience associated with different levels of T5 publications.[31]

[30]See Online Appendix Section 3.3 for results and details on specification used.

[31]The corresponding marginal effects are presented under the department rank-specific columns of Online Appendix Table O-A13.

The length of the first spell varies by individual.[32] Predictions for each rank group is obtained by estimating logit models for subsamples of faculty who had their first spell of tenure-track experience at a department within the rank group in question. For all empirical models estimated in this paper we include departmental fixed effects and adjust standard errors for clustering at the department level.

Figure 4: Predicted Probabilities for Tenure Receipt in the First Spell of Tenure-Track Employment, By Department Rank (Logit)



Note: This figure plots the predicted probabilities associated with different levels of publications in different journal categories. The predicted probability is defined in Equation TA-2 (Equation TA-2 uses parameter estimates from Equation TA-1). Department rank-specific estimates are obtained by restrictively estimating Equation TA-1 over subsamples of faculty belonging to the department rank group in question. White diamonds on the bars indicate that the prediction is significantly different than zero at the 5% level.

The figure reveals heterogeneity in the associated impact of each T5 publication in the probability of receiving tenure. Faculty at lower ranked departments face higher probabilities of tenure receipt with the same number of T5 publications. An individual with one T5 publication is predicted to face a probability of tenure of 0.2 in a top 10 department, but the same individual experiences probabilities of 0.29 and 0.39 at departments ranked 11–20 and 21–35 respectively. Faculty with two and three T5 publications at departments ranked

[32] We also estimate models that fix duration to 7 years of tenure-track experience. Pooled estimates are presented in Online Appendix Figure O-A7. The results of that analysis are qualitatively similar to the analysis in the main text. Department rank-specific estimates for tenure by the 7^{th} year are presented in Online Appendix Figures O-A8–O-A10.

11–20 are similarly predicted to experience higher probabilities of tenure than individuals in top 10 departments who have published the same number of T5 articles.[33]

2.2.2 The Power of the T5 By Quality of T5 Publications [34]

This section investigates the staying power of the T5. Results from the previous sections show that T5 publications have a powerful impact on tenure decisions, after controlling for differences in the quality of publication portfolios as proxied by citation performance of published articles. These findings suggest that the T5 influence operates through channels that are independent of publication quality. Figure 5 presents evidence in support of this hypothesis. The figure bins faculty into four quartiles based on average citations accrued through 2018 by all journal articles published by authors during the first spell of tenure-track employment. Bins are designated in the natural order of citation quality from lowest (bin 1) to highest (bin 4). Probabilities of tenure associated with different levels of T5 publications are presented within each quartile.[35] To investigate the staying power of T5 publications conditional on article quality, we require all publications to accrue citations over a minimum of ten years.[36] The analysis in this section does not adjust for departmental fixed effects and differences in the tenure process by department rank due to sample size issues. We lose a large number of observations due to restriction of the sample to individuals who completed their first spells by 2008.

Tenure probabilities generally increase with number of T5 publications across all quar-

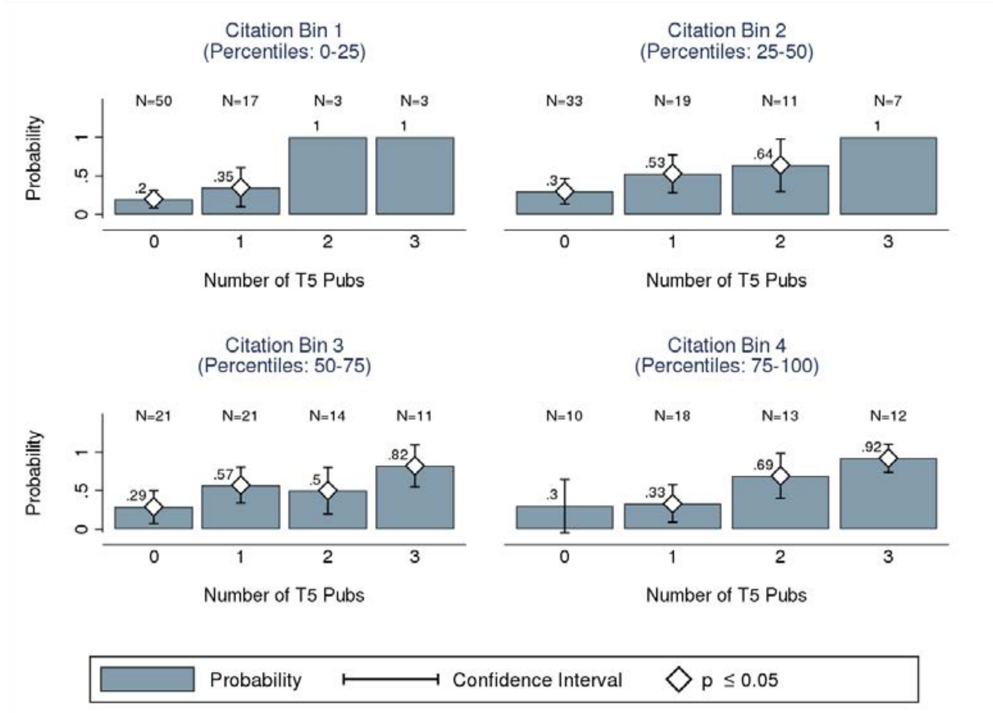
[33]While differences are evident, one cannot reject the null of equalities of the probabilities across department rank groups. See Online Appendix Table O-A14.

[34]The analysis of this section was motivated by the comments of Dan Black and Harold Uhlig.

[35]The probabilities are constructed in three steps: (i) the sample is restricted to only include faculty with 3 or more journal publications by the end of the first spell (3 is the mean number of journal publications during the first spell); (ii) each individual is binned into one of four performance quartiles based on average citations accrued through 2018 by all journal articles published by the individual during the first spell; and (iii) conditional probabilities of tenure receipt (given T5 publications) are estimated within each performance quartile by taking the proportion of individuals who received tenure given publication of zero to three T5 articles during the first spell

[36]This requirement is satisfied by restricting the estimation sample to only include individuals who completed their first spells of tenure track employment by 2008. Thus, all pre-tenure decision publications in the estimation sample must have been published in or before 2008.

Figure 5: Raw Probabilities for Receipt of Tenure in the First Spell of Tenure-Track Employment, by Quality of Overall Publications for Faculty Whose First Spell Ended by 2008 (Quality Proxied by Average Citations Received Through 2018 by First Spell Publications); Sample Restricted to Faculty With 3 or More Journal Publications by End of First Spell



Note: This figure plots estimates of tenure probabilities (by the first spell) for individuals with different numbers of T5 publications by the quality of authors' publications as proxied by citations measures through 2018. Faculty are grouped into four quartiles based on average citations accrued through 2018 by all publications during the first spell. The figure plots quartile-specific probabilities of tenure associated with each level of T5 publication. For each quartile, probabilities are estimated as the proportion of individuals with a given level of T5 publication who received tenure during the first spell. The estimation sample is restricted to only include individuals who published three or more journal articles during the first spell. Confidence intervals are not plotted for probability estimates that equal one since tenure was received by every individual within the group in question.

tiles of author publication quality. Inter-quartile comparison of tenure probabilities reveals the extent of the T5 influence. It is more valuable to have a mediocre publication portfolio with T5 publications than an outstanding portfolio without any T5s. Individuals with top quartile T5-less publication portfolios composed of three or more non-T5 publications are estimated to face similar or lower probabilities of tenure receipt than individuals with bottom quartile publication portfolios consisting of one T5 article and two or more non-T5 articles. Faculty with bottom quartile portfolios composed of two or three T5 publications have vastly greater tenure probabilities than faculty with top quartile portfolios that lack T5